

Rao Rizwan

Backend Engineer

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SUMMARY

Backend engineer with 2+ years building production Python systems. Shipped Django REST Framework APIs serving 100,000+ requests per day, moved long-running work off the request path with Celery and RabbitMQ, and tuned PostgreSQL and MongoDB for heavy throughput. Delivered containerized services on AWS with automated security scanning built into CI/CD.

PROFESSIONAL EXPERIENCE

Broadstone Technologies Apr 2026 – Sep 2026
Software Engineer Lahore, Pakistan

- Built and maintained backend REST APIs in Flask, covering endpoint design, integration, and production support.
- Integrated automated security scanning into CI/CD pipelines, surfacing vulnerabilities and risk findings across application codebases before release.
- Generated AI bill of materials (AIBOM) inventories in-pipeline, giving visibility into AI model and third-party dependency components across projects.
- Automated build, test, and release stages on AWS CodeCommit, CodePipeline, and GitHub Actions, and provisioned EC2, S3, and IAM under least-privilege access.
- Implemented OAuth 2.0 service-to-service authorization across AWS, Azure DevOps, Google Cloud, and GitHub integrations.

Programmers Force Jun 2024 – Apr 2026
Backend Engineer Lahore, Pakistan

- Designed and shipped RESTful APIs with Django REST Framework serving more than 100,000 daily requests in production.
- Introduced asynchronous task processing with Celery and RabbitMQ, moving scraping and report generation off the request path onto background workers.
- Tuned the data layer across PostgreSQL and MongoDB through schema design and query optimization to sustain heavy read and write throughput.
- Refactored core API logic and error handling, standardizing response formats and cutting production error rates.
- Containerized services with Docker and Docker Compose, and built resilient web scraping pipelines with BeautifulSoup4 and Selenium.

TECHNICAL SKILLS

Languages: Python, SQL
Backend: Django, Django REST Framework, Flask, FastAPI, REST API design, microservices, OAuth 2.0, JWT, Swagger/OpenAPI
Databases: PostgreSQL, MongoDB, Redis, schema design, query optimization
Async: Celery, RabbitMQ, background task processing, scheduled jobs
Cloud & DevOps: AWS (EC2, S3, IAM, CodePipeline, CodeCommit), Docker, Docker Compose, GitHub Actions, Azure DevOps, Google Cloud
Security: Automated vulnerability scanning, software supply chain, AI bill of materials (AIBOM), CycloneDX
Tools & Platforms: Linux, Git, pytest, Postman, BeautifulSoup4, Selenium, React

PROJECTS

SiteScopia | Website Analysis Platform FastAPI, Python, React, Vite
🌐 [Live Site](#) | 📄 [Source Code](#)

- Built a FastAPI backend that validates and fetches a public URL, parses it into structured facts, and runs independent SEO, accessibility, security, and performance analyzers as background jobs polled by a React frontend.
- Hardened the fetch path against SSRF with URL validation, rate limiting, bounded concurrency, and per-job access tokens; enriched reports via RDAP and DNS. Deployed on Render and Vercel.

JobHarvester | Job Aggregation Platform Django REST, PostgreSQL, React
🌐 [Live Site](#) | 📄 [Source Code](#)

- Built the Django REST API and PostgreSQL data model behind a live platform serving 1,300+ job listings, with search and filtering by keyword, experience level, and job type.
- Engineered scheduled crawlers with BeautifulSoup4 and Selenium to ingest postings from company career pages, and deployed end to end on Render, Vercel, and Supabase.

BOMWatch | AI Bill of Materials Scanner (in progress) FastAPI, SQLAlchemy, PostgreSQL
📄 [Source Code](#)

- Building a service that connects repositories through a GitHub App, injects a scanning workflow by pull request, and ingests the resulting CycloneDX AI-BOM artifact over a webhook for dashboard reporting, so source code never leaves GitHub.

EDUCATION

Lahore Garrison University | Bachelor of Software Engineering Lahore, Pakistan | 2020 – 2024

LEADERSHIP AND ACTIVITIES

- **Campus Director, AI Community of Pakistan (AICP):** Led campus AI awareness initiatives and technology events.
- **Participant, Aspire Leaders Program (ALP):** Selected for a globally competitive leadership program.